

Linear Programming Report

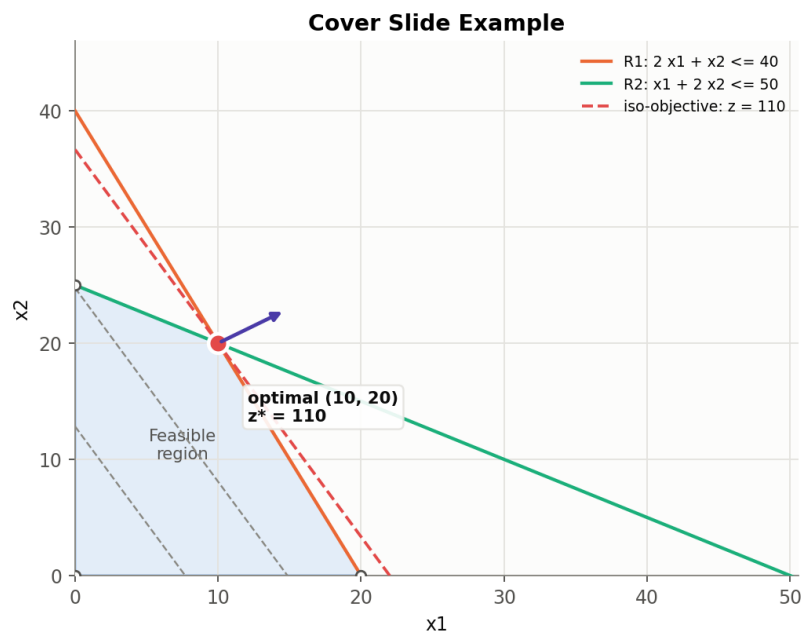
Sensitivity Analysis & Duality

Cover Slide Example

The LP pictured on the title slide: maximize $5x_1 + 3x_2$ subject to two resource constraints. Its optimal vertex is (10, 20).

Source: Lecture slide 1 (iso-value objective lines)

Problem type	MAX, 2 variables, 2 constraints
Status	OPTIMAL
Optimal Objective value	110
Dual status	OPTIMAL
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Course	Sensitivity Analysis and Duality (Dr. Islam Ali)
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1. Model formulation

MAX $z = 5x_1 + 3x_2$
subject to
 $2x_1 + x_2 \leq 40$ (R1)
 $x_1 + 2x_2 \leq 50$ (R2)
 $x_1 \geq 0, x_2 \geq 0$

Data table

Constraint	x1	x2	Sense	RHS
R1	2	1	$\leq$	40
R2	1	2	$\leq$	50
Objective c	5	3	max	—

Definitions

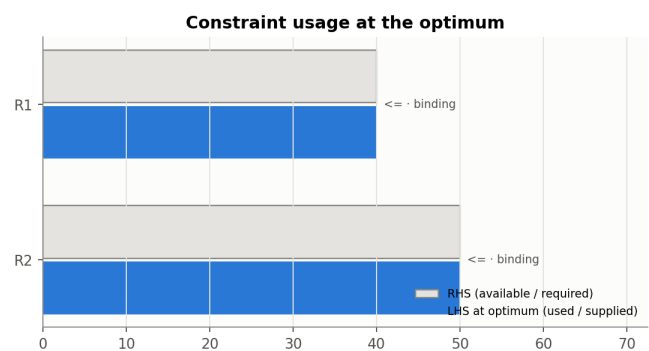
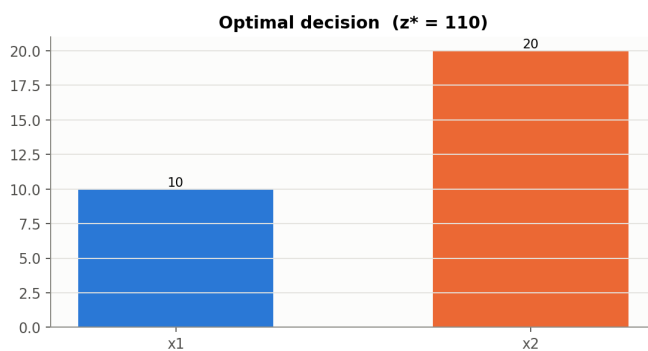
- **x1**: decision variable 1
- **x2**: decision variable 2
- **R1**: resource 1
- **R2**: resource 2

2. Optimal solution

Optimal Objective value: **110**

Variable	Description	Optimal value	Obj. coefficient	Contribution
x1	decision variable 1	10	5	50
x2	decision variable 2	20	3	60

Constraint	LHS	Sense	RHS	Slack/Surplus	Binding
R1	40	<=	40	0	yes
R2	50	<=	50	0	yes



Corner points

x1	x2	Objective value	Optimal?
0	0	0	
20	0	100	
10	20	110	✓
0	25	75	

3. Sensitivity analysis

Objective-coefficient ranging (variable cells)

Variable	Value	Reduced cost	Obj. coef	Allow. +	Allow. –	Low	High
x1	10	0	5	1	3.5	1.5	6
x2	20	0	3	7	0.5	2.5	10

Right-hand-side ranging (constraints)

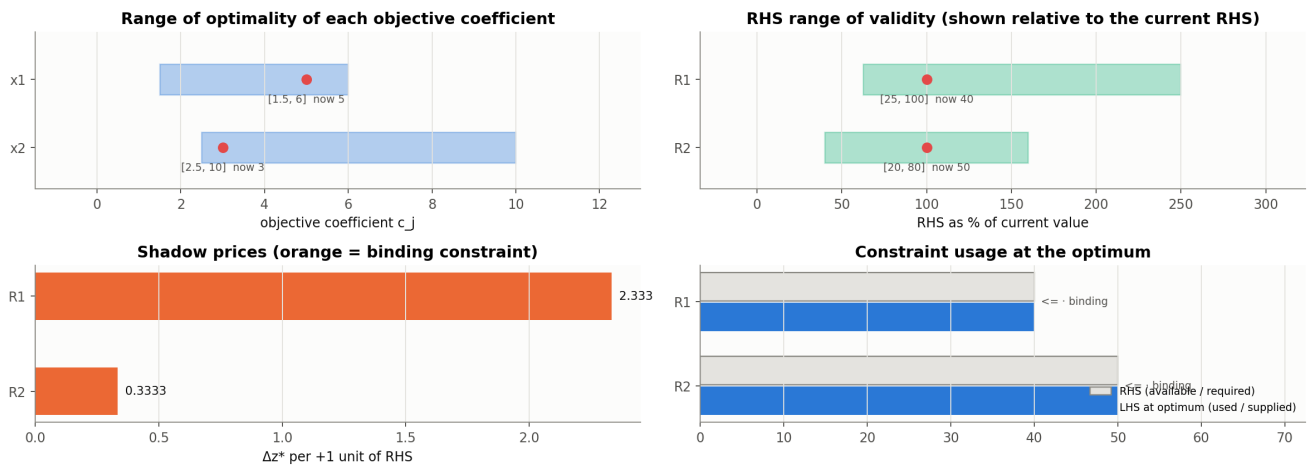
Constraint	LHS	Sense	RHS	Slack	Shadow price	Allow. +	Allow. –	Low	High
R1	40	<=	40	0	2.333	60	15	25	100
R2	50	<=	50	0	0.3333	30	30	20	80

Interpretation

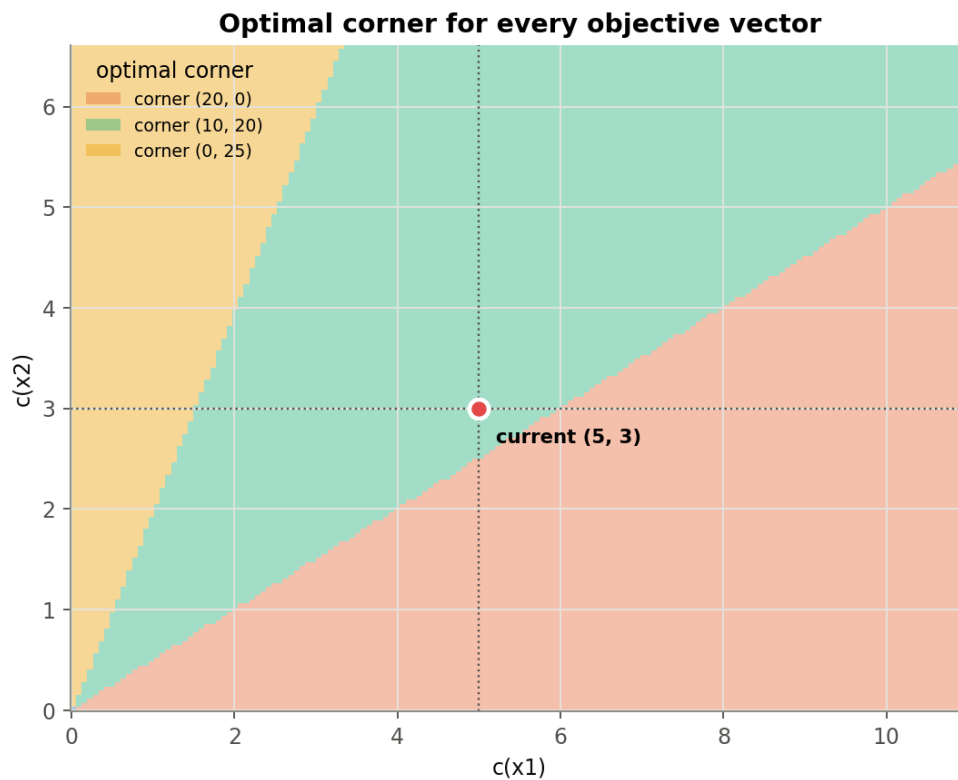
- **x1** = 10. Its objective coefficient (5) may vary in [1.5, 6] without changing the optimal solution (one change at a time).
- **x2** = 20. Its objective coefficient (3) may vary in [2.5, 10] without changing the optimal solution (one change at a time).
- **R1** is binding. Shadow price 2.333: each extra unit of RHS changes Objective value by 2.333, valid while the RHS stays in [25, 100].
- **R2** is binding. Shadow price 0.3333: each extra unit of RHS changes Objective value by 0.3333, valid while the RHS stays in [20, 80].

Sensitivity dashboard

Cover Slide Example: sensitivity dashboard ($z^* = 110$)

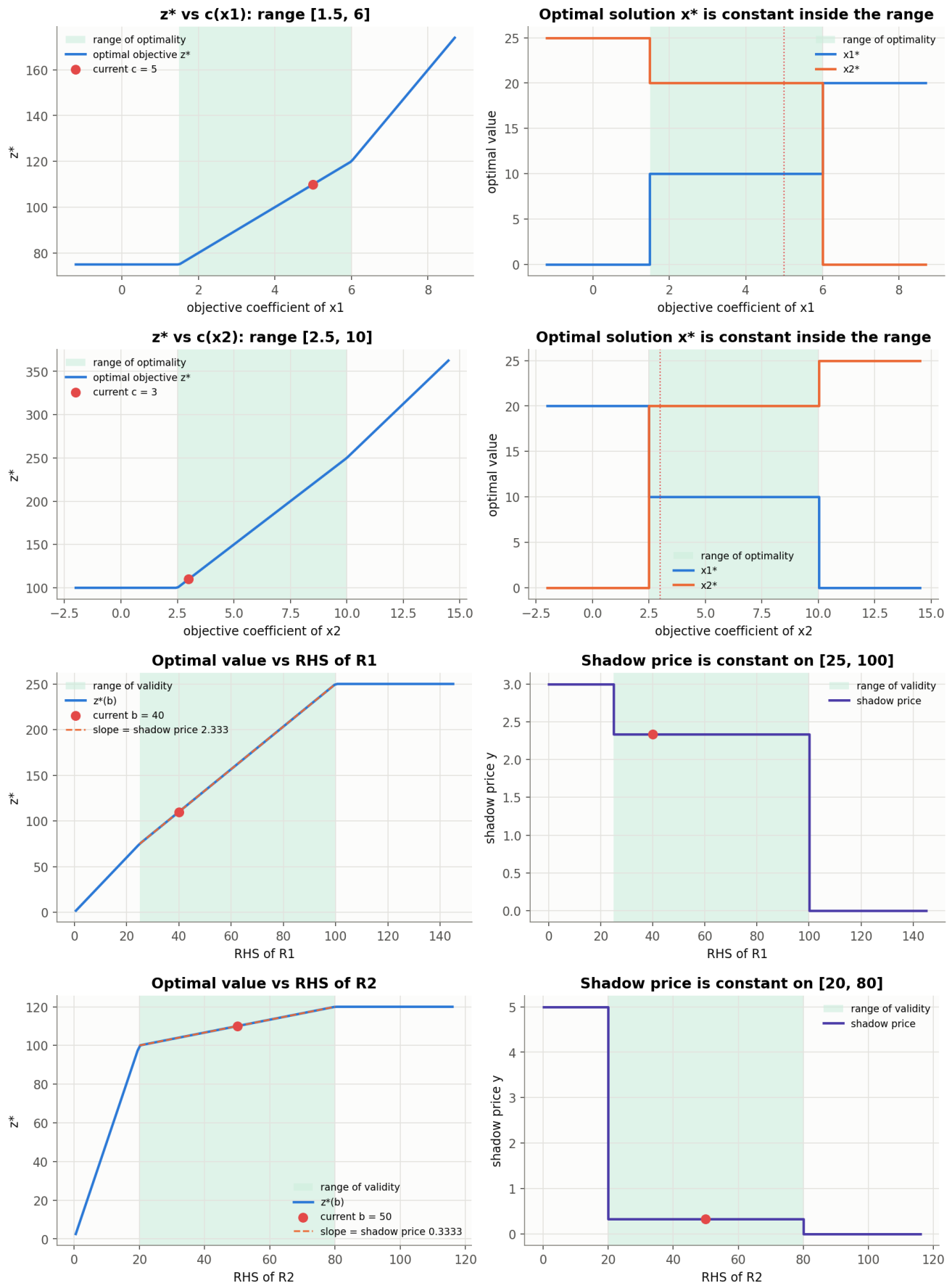


Optimal corner for every objective vector



4. Parametric analysis

Each chart re-solves the LP for a range of values of one coefficient. The shaded band is the range reported in section 3.

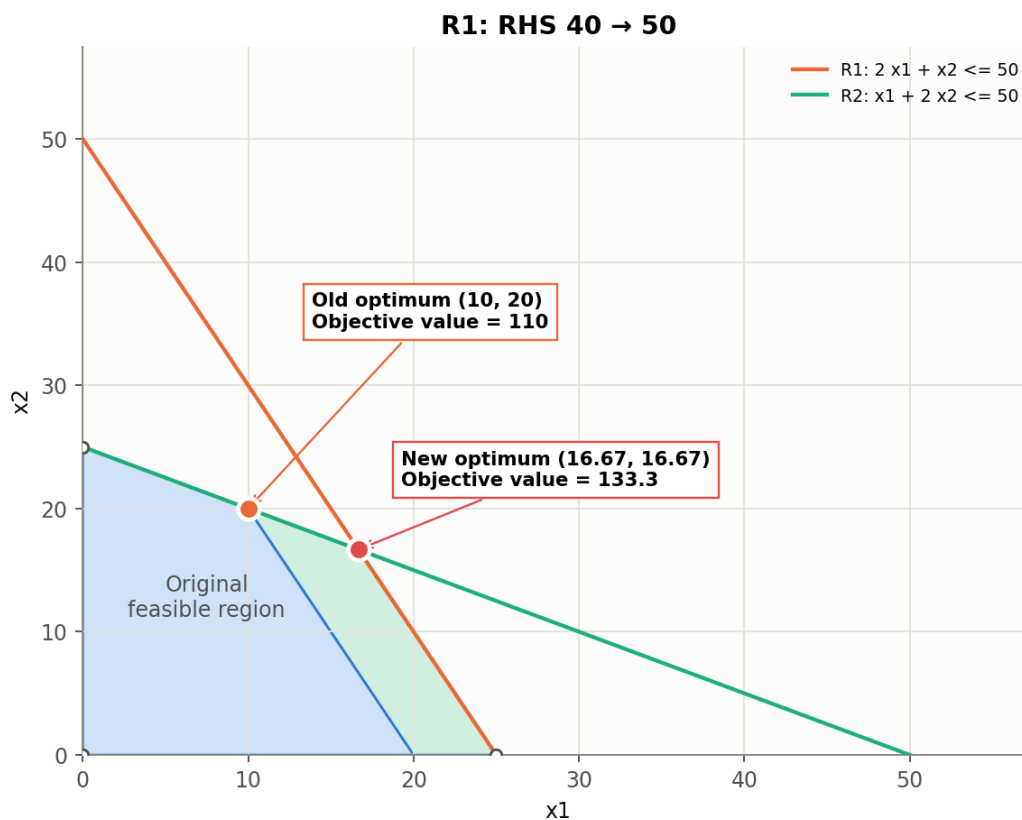
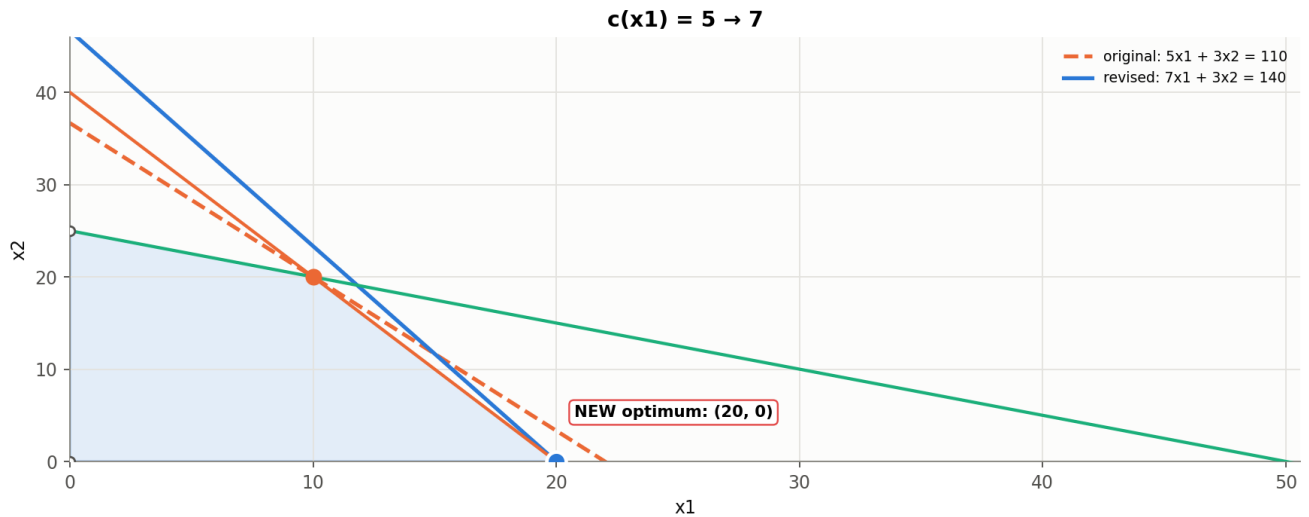


5. What-if scenarios

'Predicted' uses only the sensitivity report (shown when the change is inside its range); 'Actual' re-solves the LP.

Scenario	InRange	Predicted	Actual	NewSolution	Note
c(x1): 5 → 7	no	—	140	20, 0	steeper objective
b(R1): 40 → 50	yes	133.3	133.3	16.67, 16.67	more of resource 1

Changing an objective coefficient rotates the iso-profit line; the feasible region never moves



6. Duality

Dual formulation

MIN $z = 40 y_{R1} + 50 y_{R2}$
 subject to
 $2 y_{R1} + y_{R2} \geq 5$ (x1)
 $y_{R1} + 2 y_{R2} \geq 3$ (x2)
 $y_{R1} \geq 0, y_{R2} \geq 0$

Dual solution (resource / requirement prices)

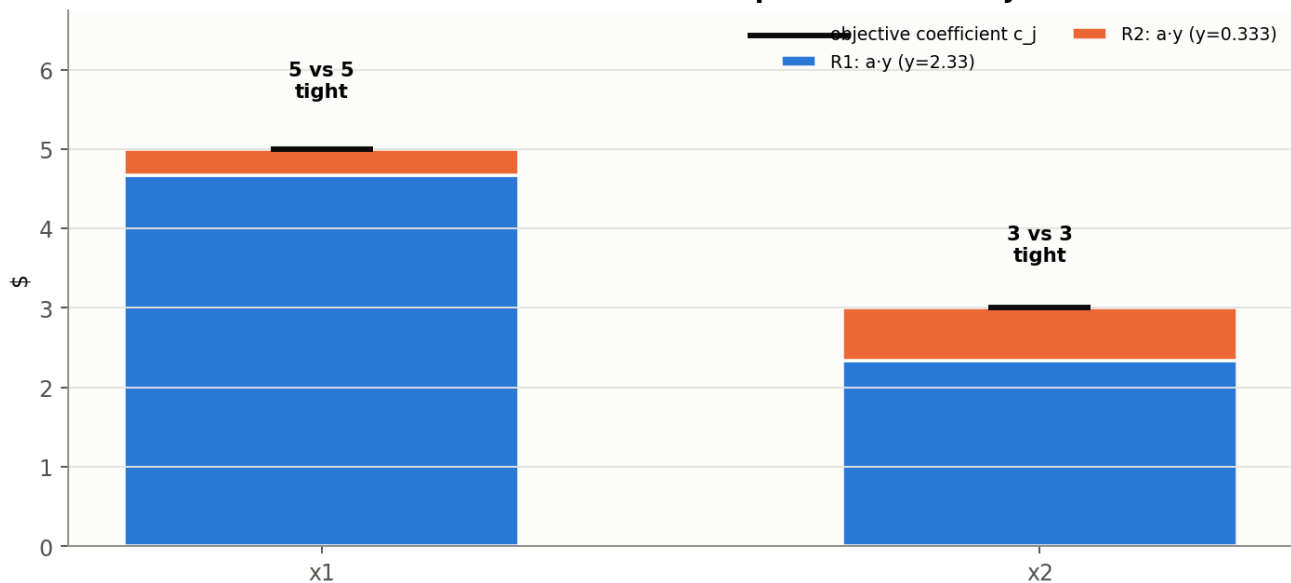
Dual variable	Primal constraint	Value y^*	Primal shadow price	$b_i \cdot y_i$
y_{R1}	R1	2.333	2.333	93.33
y_{R2}	R2	0.3333	0.3333	16.67

Strong duality check: primal $z^* = 110$, dual $w^* = 110$ (equal ✓). Dual of the dual equals the primal: yes.

Complementary slackness (all products must be 0)

Pair	Slack	DualValue	Product
R1 slack $\times y_{R1}$	0	2.333	0
R2 slack $\times y_{R2}$	0	0.3333	0
$x1 \times$ surplus of dual constraint x1	0	10	0
$x2 \times$ surplus of dual constraint x2	0	20	0

Dual constraints: value of resources in each product vs its objective coefficient

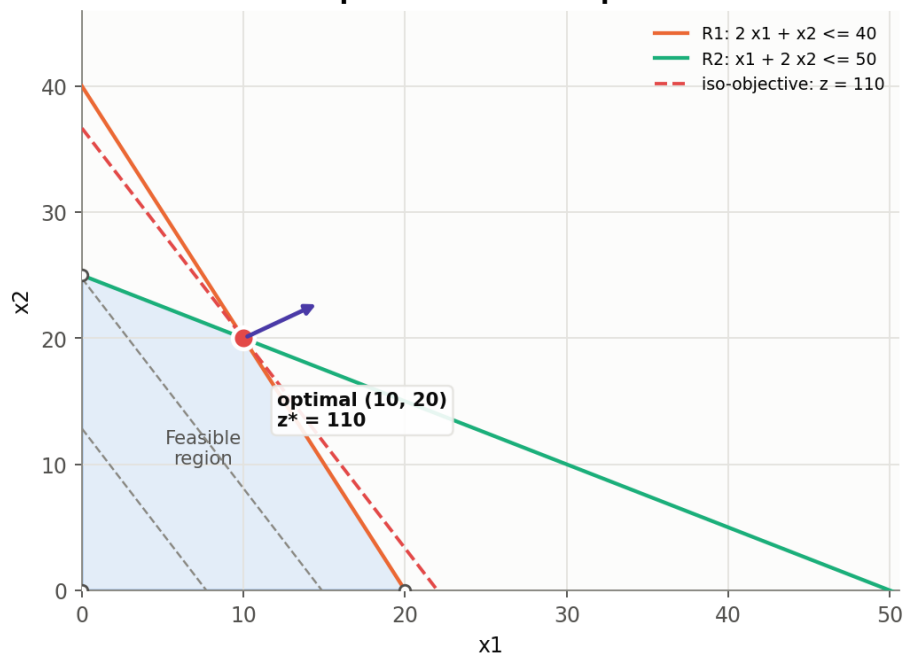


Primal (max): 2 constraints × 2 variables

5	3	← objective
2	1	
1	2	≤ 40
		≤ 50
x_1	x_2	

Dual (min): 2 constraints × 2 variables

40	50	← objective
2	1	
1	2	≥ 5
		≥ 3
y_{R1}	y_{R2}	

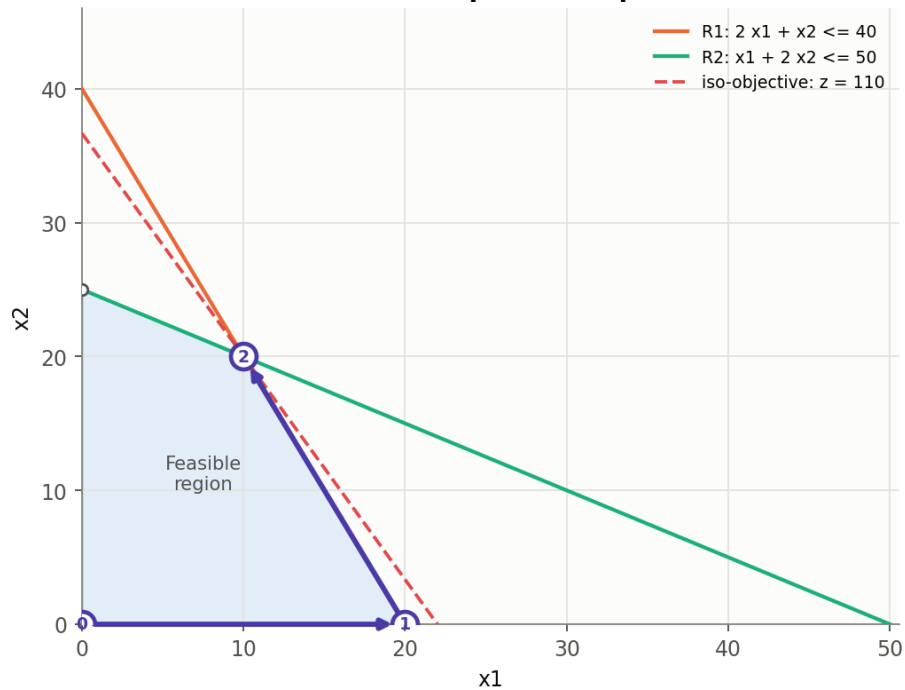
Graphical view of the primal

7. Simplex iterations

Two-phase tableau simplex (entering column = most negative $z_i - c_i$; leaving row = minimum ratio).

Tableau	Method	Phase	Entering	Leaving	Objective	PrimalFeasible	DualFeasible	Point
0	primal	2	x1	s1	0	yes	no	(0, 0)
1	primal	2	x2	s2	100	yes	no	(20, 0)
2	primal	2			110	yes	yes	(10, 20)

Path of the primal simplex



Primal simplex - phase 2, tableau 0
enter x1, leave s1 (min ratio 20)

	x1	x2	s1	s2	RHS
z	-5	-3	0	0	0
s1	2	1	1	0	40
s2	1	2	0	1	50

Primal simplex - phase 2, tableau 1
enter x2, leave s2 (min ratio 20)

	x1	x2	s1	s2	RHS
z	0	-0.5	2.5	0	100
x1	1	0.5	0.5	0	20
s2	0	1.5	-0.5	1	30

Primal simplex - phase 2, tableau 2
optimal tableau

	x1	x2	s1	s2	RHS
z	0	0	2.33	0.333	110
x1	1	0	0.667	-0.333	10
x2	0	1	-0.333	0.667	20